

LAB 13b — Enterprise Knowledge Base Ingestion with OneDrive & Vector Store

Objective

Build an enterprise AI knowledge ingestion workflow in n8n that:

- retrieves enterprise documents from OneDrive
- loops through multiple files automatically
- downloads enterprise documents dynamically
- extracts document content
- converts enterprise knowledge into embeddings
- stores embeddings inside a vector database
- prepares AI-searchable enterprise knowledge
- connects the vector database to an AI Agent for retrieval

This lab introduces a complete enterprise ingestion + retrieval architecture for AI-powered knowledge systems.

Skills

- OneDrive automation
- Enterprise ingestion workflows
- Loop processing
- Vector databases
- Embeddings
- AI Agent tools
- Enterprise RAG architecture
- Knowledge retrieval systems
- AI workflow orchestration

Enterprise Scenario

A manufacturing company stores SOPs, troubleshooting guides, maintenance procedures, and operational manuals inside OneDrive.

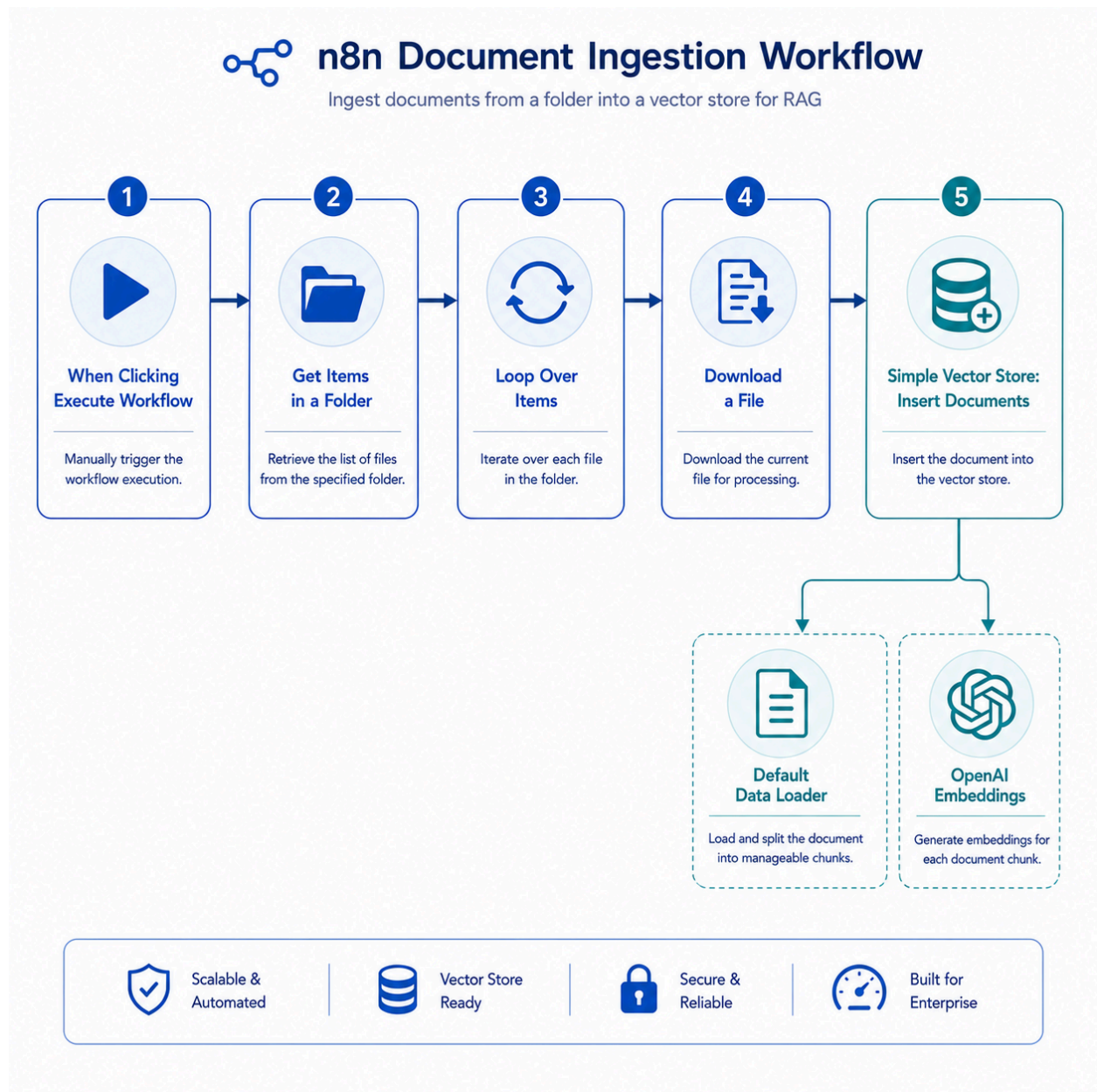
The enterprise AI system should:

- automatically discover documents
- process multiple enterprise files

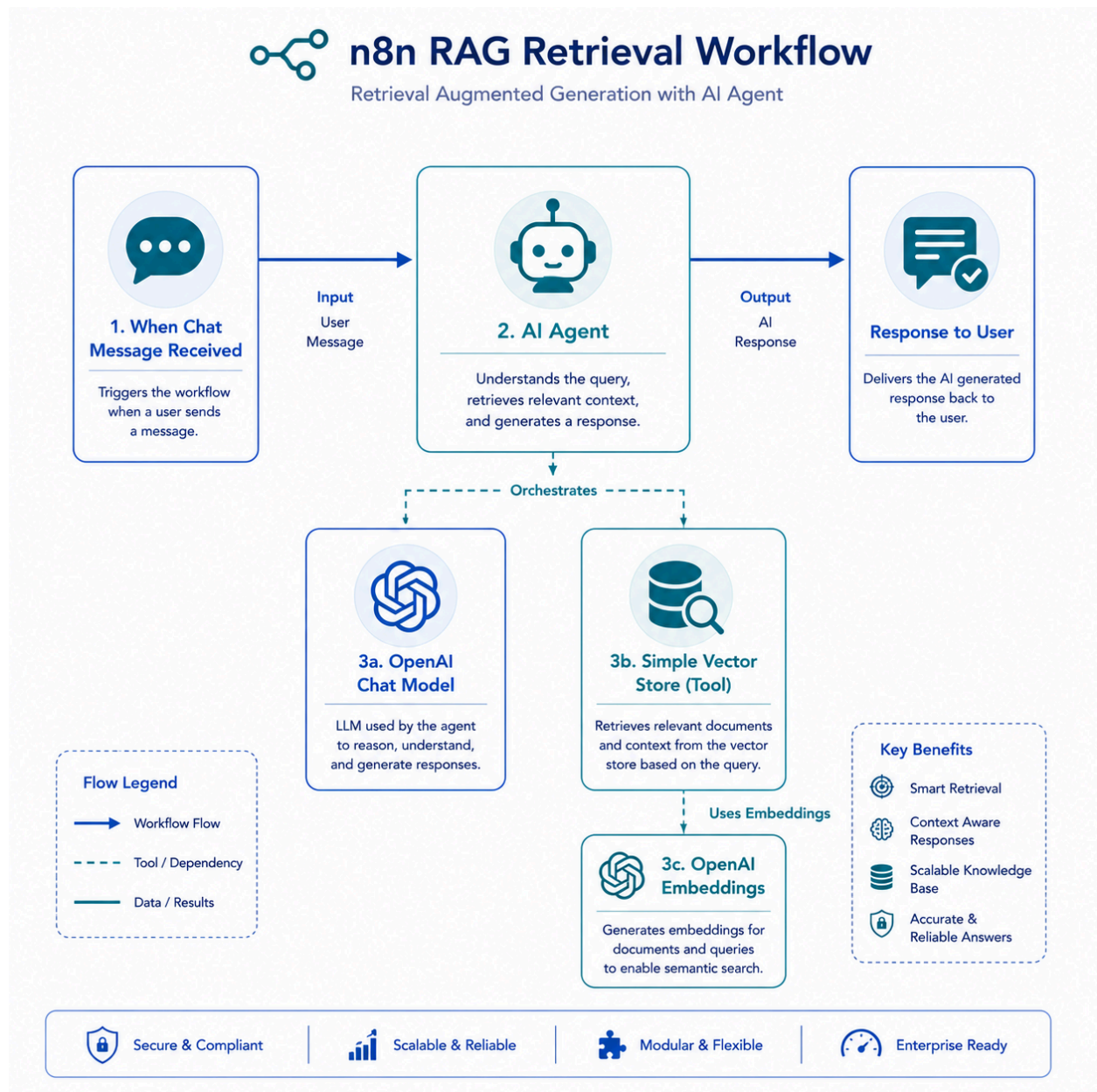
- build searchable AI knowledge
- allow AI Agents to retrieve enterprise knowledge dynamically

Architecture Overview

INGESTION FLOW



RETRIEVAL FLOW



Step-by-Step with Code

Step 1 — Create Workflow

Workflow name:

Enterprise Knowledge Base Pipeline

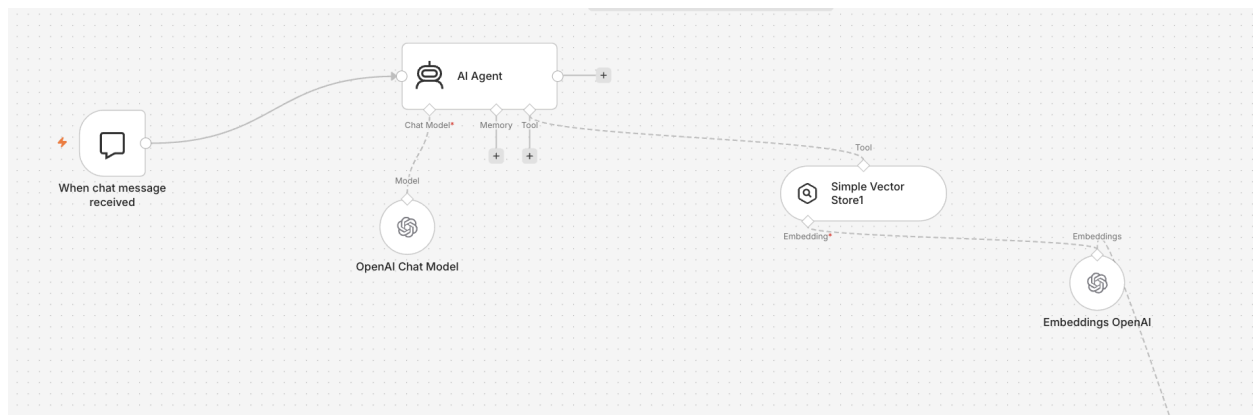
Step 2 — Create Retrieval Chat Flow

Add nodes:

- When Chat Message Received
- AI Agent
- OpenAI Chat Model
- Simple Vector Store
- Embeddings OpenAI

Purpose:

allow users to chat with enterprise knowledge



Step 3 — Configure OpenAI Chat Model

Add:

OpenAI Chat Model

Example model:

gpt-4.1-mini

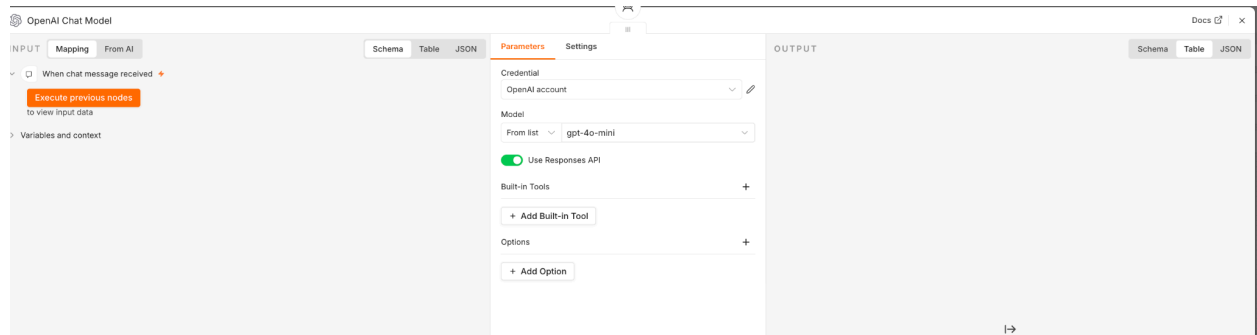
Connect:

OpenAI Chat Model

→ AI Agent (Chat Model)

Purpose:

provide reasoning capability for enterprise AI assistant



Step 4 — Configure Vector Store Retrieval Tool

Add:

Simple Vector Store

Mode:

Retrieve Documents (Tool)

Connect:

Embeddings OpenAI

→ Simple Vector Store

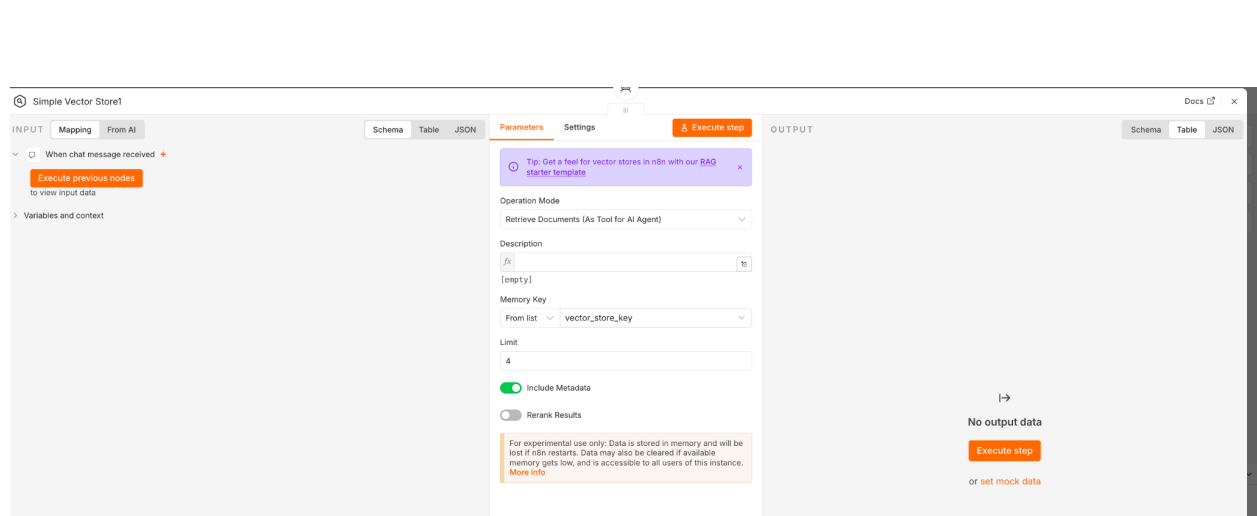
Then connect:

Simple Vector Store

→ AI Agent (Tool)

Purpose:

allow AI Agent to search enterprise knowledge dynamically



Step 5 — Configure Enterprise AI System Prompt

Inside AI Agent:

You are an enterprise manufacturing AI assistant.

Use the connected knowledge base to answer questions.

Always:

- retrieve relevant enterprise knowledge
- provide concise operational guidance
- prioritize SOP and maintenance procedures
- avoid hallucinating unsupported information

Purpose:

enforce enterprise-safe AI responses

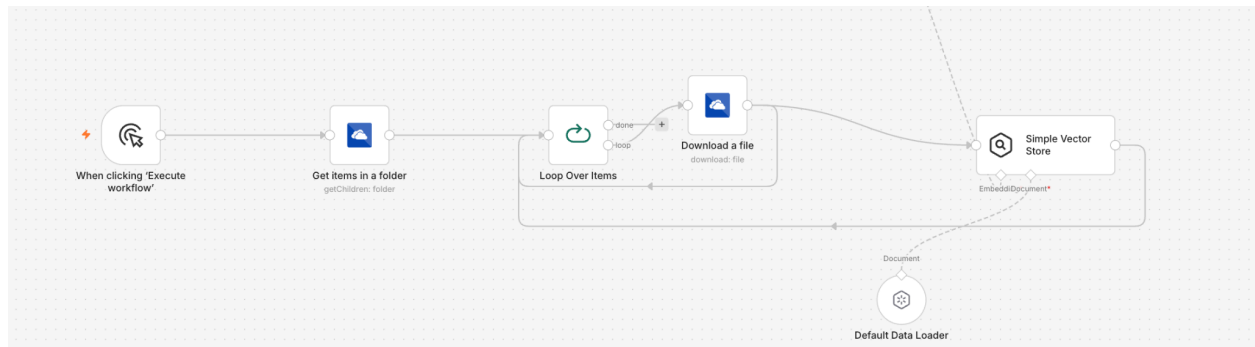
Step 6 — Create Enterprise Ingestion Flow

Add nodes:

- When clicking 'Execute workflow'
- Microsoft OneDrive
- Loop Over Items
- Download a File
- Simple Vector Store
- Default Data Loader
- Embeddings OpenAI

Purpose:

build enterprise knowledge base automatically



Step 7 — Configure OneDrive Folder Retrieval

Configure Microsoft OneDrive node:

Operation:

Get Items in Folder

Example folder:

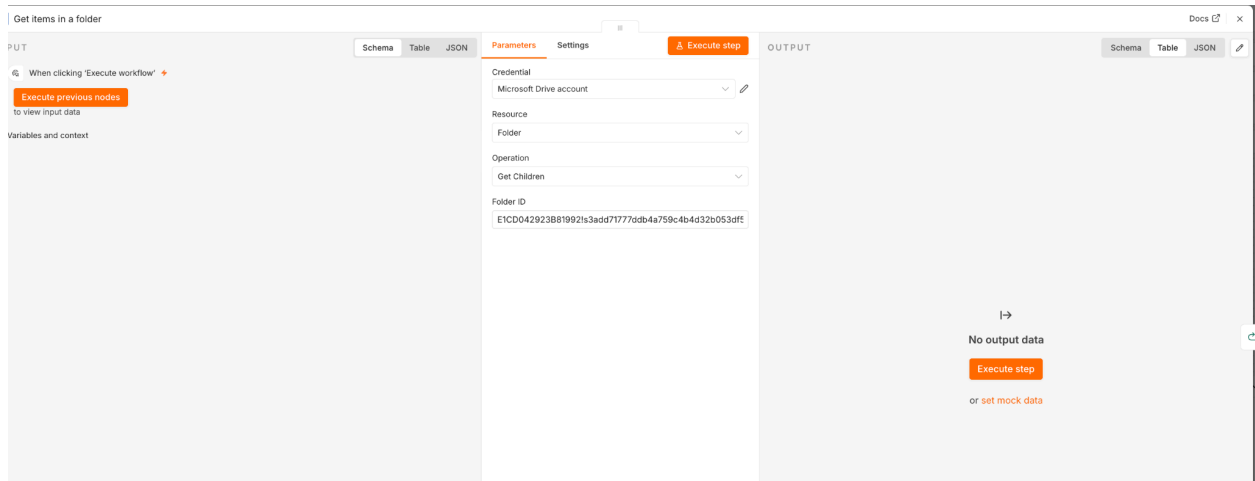
Manufacturing_SOPs

Purpose:

retrieve enterprise documents automatically

Enterprise examples:

- SOP manuals
- maintenance guides
- troubleshooting procedures
- operational documentation



Step 8 — Configure Loop Processing

Add:

Loop Over Items

Connect:

OneDrive

→ Loop Over Items

Purpose:

process multiple enterprise documents dynamically

Enterprise pattern:

OneDrive Folder

→ Multiple Files

→ Automated AI Knowledge Pipeline

Step 9 — Configure File Download

Add:

Download a File

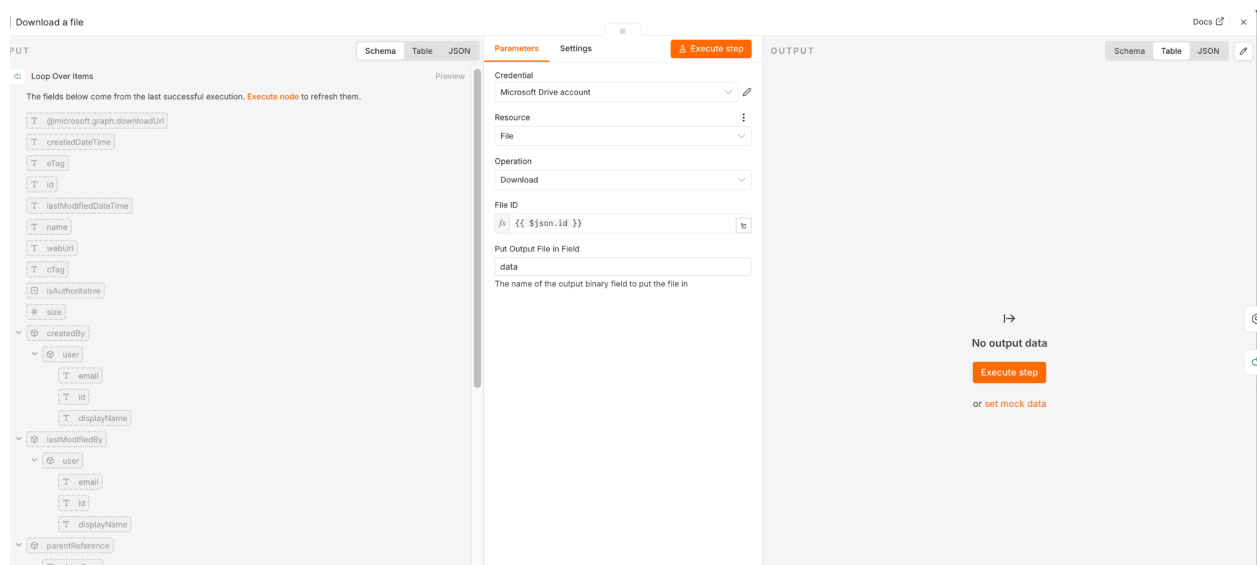
Connect:

Loop Over Items

→ Download a File

Purpose:

retrieve actual enterprise document contents



Step 10 — Configure Vector Database Ingestion

Add:

Simple Vector Store

Mode:

Insert Documents

Connect:

Download a File

→ Simple Vector Store

Then connect:

Default Data Loader

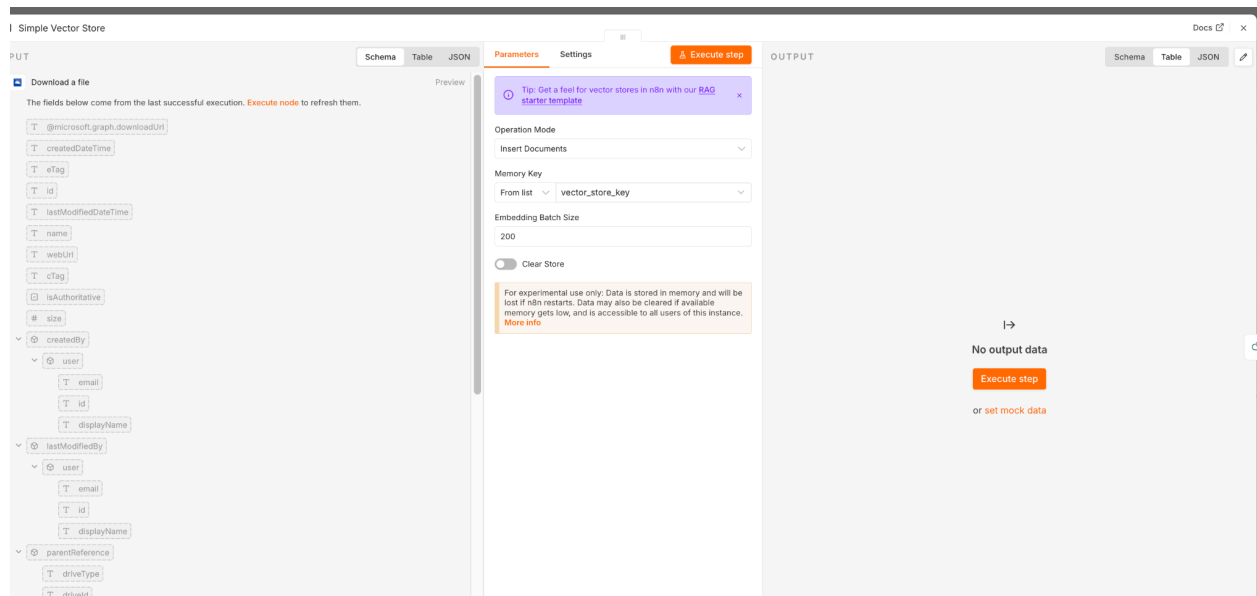
→ Simple Vector Store (Document Input)

Embeddings OpenAI

→ Simple Vector Store (Embedding Input)

Purpose:

convert enterprise documents into searchable AI knowledge



Step 11 — Configure Default Data Loader

Add:

Default Data Loader

Connect:

Default Data Loader

→ Simple Vector Store

Purpose:

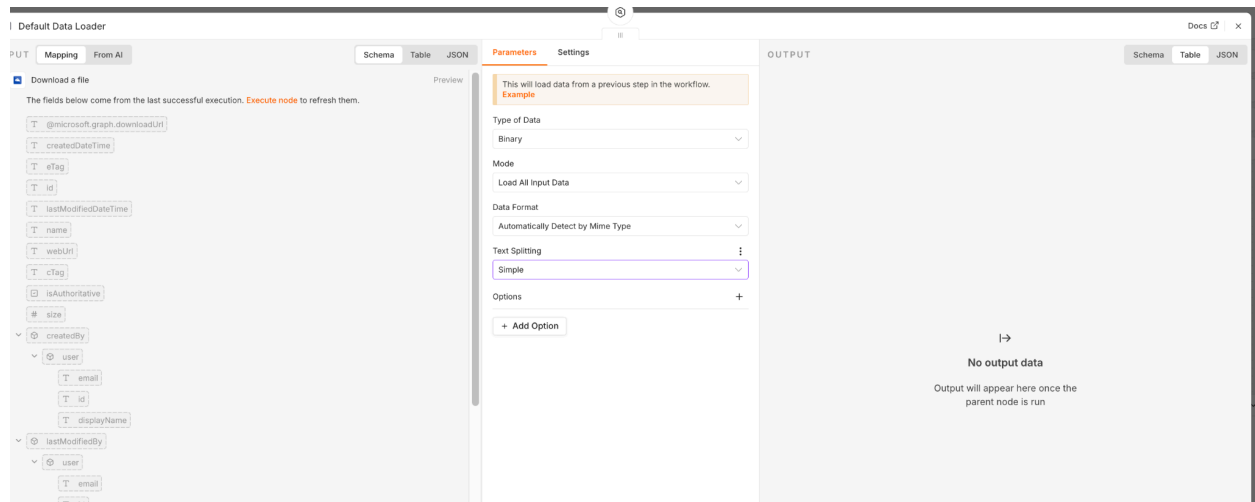
automatically extract and prepare enterprise document content

Supported files:

- TXT
- PDF
- DOCX

Purpose:

simplify enterprise ingestion pipeline without manual routing



Step 12 — Configure OpenAI Embeddings

Add:

Embeddings OpenAI

Example model:

text-embedding-3-small

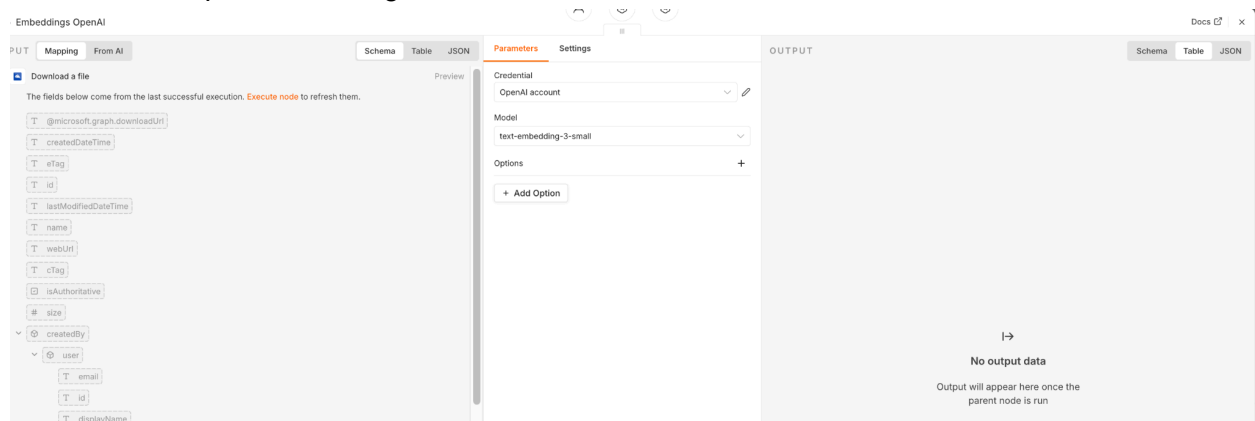
Connect:

Embeddings OpenAI

→ Simple Vector Store

Purpose:

transform enterprise knowledge into searchable vectors



Step 13 — Complete Loop Connection

Connect:

Simple Vector Store

→ Loop Over Items

Purpose:

continue processing remaining enterprise files automatically

Step 14 — Execute Enterprise Ingestion Workflow

Run workflow.

Expected ingestion flow:

OneDrive Files

→ Loop Processing

→ File Download

→ Document Loading

→ Embeddings

→ Vector Database Storage

Step 15 — Test Enterprise AI Retrieval

Open chat interface.

Example prompts:

How do I restart the cooling system?

What is the maintenance checklist for conveyor inspection?

Show SOP steps for production shutdown.

Expected behavior:

- AI retrieves enterprise knowledge
 - AI uses vector database context
 - AI responds using uploaded SOP documents
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Example Enterprise Documents

Example files:

- cooling_system_sop.pdf
 - maintenance_checklist.docx
 - login_issue_guide.txt
 - production_deviation_manual.pdf
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Expected Workflow Output

- enterprise files discovered successfully
 - multiple documents processed automatically
 - embeddings generated successfully
 - enterprise vectors stored successfully
 - AI Agent retrieves enterprise knowledge dynamically
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Final Workflow Architecture

Retrieval Layer

When Chat Message Received

→ AI Agent

└─ OpenAI Chat Model

└─ Simple Vector Store (Retrieve Tool)

└─ Embeddings OpenAI

Ingestion Layer

When Clicking Execute Workflow

→ Get Items in Folder

→ Loop Over Items

→ Download a File

→ Simple Vector Store (Insert Documents)

└─ Default Data Loader

└─ Embeddings OpenAI

Why This Lab Matters

This lab introduces a complete enterprise AI knowledge architecture.

Participants learn:

- enterprise AI ingestion pipelines
- vector database workflows
- enterprise document automation
- AI retrieval systems
- RAG architecture
- scalable enterprise AI patterns
- AI Agent tool integration

This architecture is commonly used in:

- enterprise copilots
- manufacturing AI assistants
- AI-powered SOP systems
- internal knowledge assistants
- enterprise RAG platforms
- MCP-enabled enterprise AI systems

Enterprise AI Pattern Introduced

Enterprise Documents

- Automated Discovery
- Loop Processing
- AI Embeddings
- Vector Database
- AI Agent Retrieval
- Enterprise Knowledge Responses

Optional Exercise

Enhance workflow with:

- scheduled ingestion
- multi-folder processing
- metadata tagging
- document categorization
- department-based filtering
- audit logging
- confidence scoring
- document version tracking

Example metadata:

```
{  
  "document_type": "SOP",  
  "department": "Manufacturing",  
  "source": "OneDrive",  
  "version": "v2.1"  
}
```

Optional Advanced Exercise — Production Vector Database

Replace:

Simple Vector Store

with:

- Qdrant
- Pinecone
- Weaviate
- PostgreSQL + pgvector

Benefits:

- persistent enterprise storage
- scalable retrieval
- production-ready RAG
- survives workflow restarts
- multi-user enterprise search

Key Takeaways

- Enterprise AI systems require automated ingestion pipelines
- OneDrive supports centralized enterprise document management
- Loop processing enables scalable AI automation
- Embeddings transform enterprise knowledge into searchable vectors
- Vector databases power enterprise AI retrieval systems
- AI Agents can dynamically retrieve enterprise operational knowledge
- RAG architecture enables grounded enterprise AI responses